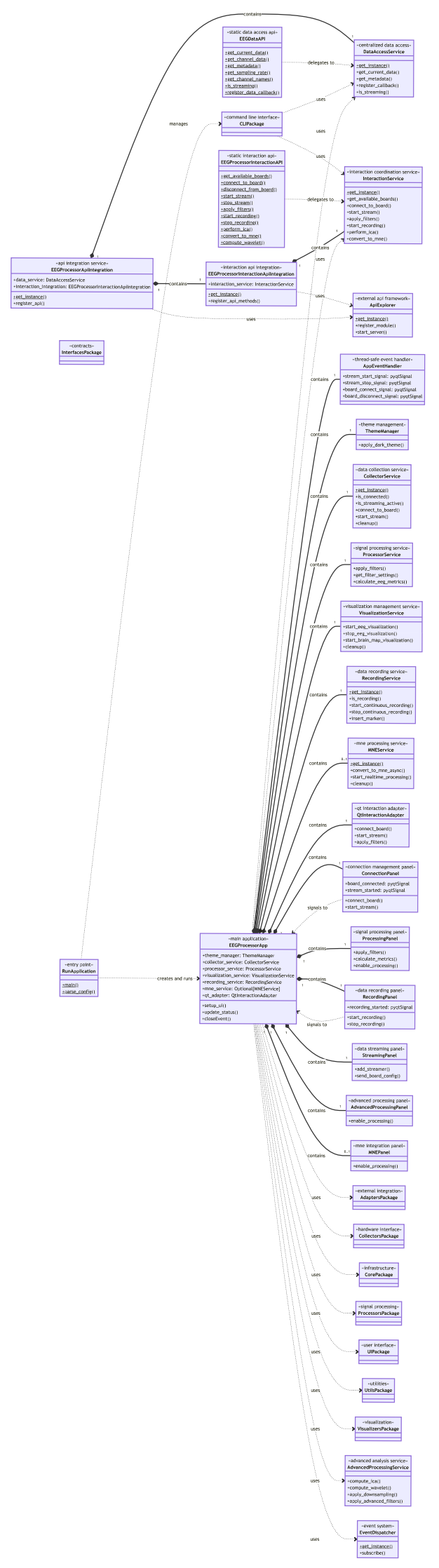

«static data access api»
EEGDataAPI
+get_current_data()
+get_channel_data()
+get_metadata()
+get_sampling_rate()
+get_channel_names()
+is_streaming()
+register_data_callback()
«centralized data access»
DataAccessService
+get_instance()
+get_current_data()
+get_metadata()
+register_callback()
+is_streaming()
delegates to
contains
uses
manages
«command line interface»
CLIPackage
«static interaction api»
EEGProcessorInteractionAPI
+get_available_boards()
+connect_to_board()
+disconnect_from_board()
+start_stream()
+stop_stream()
+apply_filters()
+start_recording()
+stop_recording()
+perform_ica()
+convert_to_mne()
+compute_wavelet()
«interaction coordination service»
InteractionService
+get_instance()
+get_available_boards()
+connect_to_board()
+start_stream()
+apply_filters()
+start_recording()
+perform_ica()
+convert_to_mne()
«api integration service»
EEGProcessorApiIntegration
+data_service: DataAccessService
+interaction_integration: EEGProcessorInteractionApiIntegration
+get_instance()
+register_api()
«interaction api integration»
EEGProcessorInteractionApiIntegration
+interaction_service: InteractionService
+get_instance()
+register_api_methods()
«external api framework»
ApiExplorer
+get_instance()
+register_module()
+start_server()
«contracts»
InterfacesPackage
«thread-safe event handler»
AppEventHandler
+stream_start_signal: pyqtSignal
+stream_stop_signal: pyqtSignal
+board_connect_signal: pyqtSignal
+board_disconnect_signal: pyqtSignal
«theme management»
ThemeManager
+apply_dark_theme()
«data collection service»
CollectorService
+get_instance()
+is_connected()
+is_streaming_active()
+connect_to_board()
+start_stream()
+cleanup()
«signal processing service»
ProcessorService
+apply_filters()
+get_filter_settings()
+calculate_eeg_metrics()
«visualization management service»
VisualizationService
+start_eeg_visualization()
+stop_eeg_visualization()
+start_brain_map_visualization()
+cleanup()
«data recording service»
RecordingService
+get_instance()
+is_recording()
+start_continuous_recording()
+stop_continuous_recording()
+insert_marker()
«mne processing service»
MNEService
+get_instance()
+convert_to_mne_async()
+start_realtime_processing()
+cleanup()
«qt interaction adapter»
QtInteractionAdapter
+connect_board()
+start_stream()
+apply_filters()
«connection management panel»
ConnectionPanel
+board_connected: pyqtSignal
+stream_started: pyqtSignal
+connect_board()
+start_stream()
signals to
«main application»
EEGProcessorApp
+theme_manager: ThemeManager
+collector_service: CollectorService
+processor_service: ProcessorService
+visualization_service: VisualizationService
+recording_service: RecordingService
+mne_service: Optional[MNEService]
+qt_adapter: QtInteractionAdapter
+setup_ui()
+update_status()
+closeEvent()
«signal processing panel»
ProcessingPanel
+apply_filters()
+calculate_metrics()
+enable_processing()
«data recording panel»
RecordingPanel
+recording_started: pyqtSignal
+start_recording()
+stop_recording()
«data streaming panel»
StreamingPanel
+add_streamer()
+send_board_config()
«advanced processing panel»
AdvancedProcessingPanel
+enable_processing()
«mne integration panel»
MNEPanel
+enable_processing()
«entry point»
RunApplication
+main()
+parse_config()
creates and runs
«external integration»
AdaptersPackage
«hardware interface»
CollectorsPackage
«infrastructure»
CorePackage
«signal processing»
ProcessorsPackage
«user interface»
UIPackage
«utilities»
UtilsPackage
«visualization»
VisualizersPackage
«advanced analysis service»
AdvancedProcessingService
+compute_ica()
+compute_wavelet()
+apply_downsampling()
+apply_advanced_filters()
«event system»
EventDispatcher
+get_instance()
+subscribe()